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Effect of submergence stress on the rice, *Oryza sativa* L. proteome

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Abstract : Water status is the main factor affecting rice production. The submergence due to flash-floods is the key factor limiting the yield of rice that grows in lowland ecosystems. During response and adaptation to the submergence stress, many stress-related genes are induced and the levels of a variety of stress resistance-related functional proteins are accumulated. The alteration of protein synthesis or degradation is one of the fundamental metabolic processes that may influence submergence tolerance. We report here a proteomic approach to investigate the effect of submergence stress on the rice (*Oryza sativa* L.) proteome. Plants were stressed for 24, 48, 72 and 96 hours. Total proteins were extracted and separated by one dimensional gel electrophoresis followed by Coomassie staining.

Keywords : Submergence stress, *Oryza sativa* L.

INTRODUCTION

A Plants are especially dependent on environmental factors, and continuous abiotic stress can have harmful effects on them. Most plants are highly sensitive to anoxia during submergence. Flash- floods are highly unpredictable and may occur at any growth stage of rice and yield loss may be less than 10 to 100%, depending on factors such as water depth, duration of submergence, temperature, turbidity of water, rate of nitrogen application, light intensity and the age of the crop **(Setter et al. 1997).** Most rice cultivars show shoot elongation in response to submergence. Elongation ability of leaves and internodes are essential to keep pace with the rising water and to escape complete submergence, thus ensuring O_2 supply and access to CO_2 and light **(Setter et al. 1997).** Flooding increases ethylene concentration in plant tissue both due to enhanced synthesis as well as entrapment. Germination is highly sensitive to flooding. During later vegetative development, rice can adapt to complete submergence through processes that provide necessary energy for maintenance metabolism and minimize losses. An important aspect of the adaptation to oxygen limitation includes metabolic changes such as avoidance of self poisoning and cytoplasmic acidosis, maintenance of adequate supplies of energy and sugar.

During the last decade, major advances have been made in plant genetic engineering **(Grover and Minhas 2000).** The methods for stable genetic transformation as well as regulation of introduced trans-genes have been optimized to a great extent. Rapid advances in genomic technologies are leading to an increased understanding of global gene expression in plants. Recently, proteomics has become an essential methodology for large scale analysis of protein in the several aspects of plant biology **(Pandey and Mann, 2000).** Proteomics means investigation of biological processes by the systematic analysis of a large number of expressed proteins for specific properties such as their identity, quantity, activity and molecular interactions **(Pandey and Mann 2000, Gygi and Abersold 2000).** Proteome analysis using differential display with two-dimensional sodium dodecyl sulphate -polyacrylamide gel electrophoresis (2-D SDS-PAGE) has several advantages over other approaches **(Pandey *et al.*, 2006).** Under

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submergence stress expression pattern of various proteins changed. Therefore, the aim of the present research was to study the changes in proteins under submergence stress in contrasting rice shoots with the use of one and two-dimensional gel electrophoresis.

MATERIALS AND METHODS

The present study was aimed at a better understanding of the molecular adaptation mechanisms of submergence stress**.** It is implicated with identifying the proteins that are specifically altered in rice (Oryza sativa L.) during submergence stress and during recovery from the stress. Seeds of submergence tolerant Swarna (Sub 1) and submergence sensitive Ratna rice type's were washed thoroughly in tap water followed by detergent for 10 min. The seeds were further treated with (30% v/v) Sodium hypochlorite solution for 30 minutes. The surface sterilized seeds were shown on absorbent cotton pads placed in plastic trays and allowed to germinate. Four day old rice seedlings were subjected to submergence stress and transferred to control condition for recovery. Stress was given for different duration of time (24, 48, 72 and 96 hours) and after that, the seedlings were recovered for the same period. Samples were harvested after each stress treatment and recovery.

Samples for 1D analysis were prepared following the procedure of **Dubey *et al.* (2003).** Shoots of rice seedlings were homogenized in liquid Nitrogen up to the formation of fine powder. Polyvinyl poly pyrrolidone (PVPP) was added to the samples (50 mg/ g fresh weight of the tissue) during homogenization. The powder was then dissolved in **Zivy *et al.* (1985)** buffer containing 30 mM Tris-Hcl (pH 8.8), 1 mM L-ascorbic acid, 1 mM EDTA-Na (Ethylenediaminetetraacetic acid), 5 mM $MgCl_{2,}$ 1 mM dithiothreitol (DTT) and 1 mM phenyl methyl sulphonyl fluoride (PMSF). This extract was centrifuged twice at 10,000 rpm for 15 min at 4°C and the supernatant was transferred to SS34 centrifuge tubes. Then the soluble protein extract was precipitated in pre-chilled acetone and â-mercaptoethanol (0.07% v/v) solution overnight at –20°C. It was again centrifuged and the pellet, which contained the total protein, was dissolved in **Laemmli (1970)** buffer containing 62.5 mM Tris-HCl (pH 6.8), 10% (v/v) glycerol, 5% (v/v) â-mercaptoethanol and 2% (w/v) sodium dodecyl sulphate (SDS). This solution was then heated in boiling water for 5 min. After centrifugation, required amount of supernatant was taken to estimate the protein concentration by **Bradford (1976)** method. Buffer soluble proteins were extracted and analyzed by 1 mm thick, vertical gels. Coomassie staining of 1D gels were performed according to **Dubey *et al.*, 2003**.

RESULTS AND DISCUSSION

We applied proteomics approach for the analysis of rice cells treated with submergence stress. Swarna (Sub 1) and Ratna rice shoot tissues were subjected to submergence stress treatments for 24, 48, 72 and 96 h in this experiment. It is possible that differences in protein profiles between uninduced control and stressed samples could partly be due to growth of seedlings. The protein changes that are associated with the development of the seedlings during the period of this experiment were taken in to consideration while documenting stress-induced protein alterations.

Most of the cellular proteins showed similar pattern of alterations in shoot tissues of rice seedlings. Several proteins were below the level of detection in uninduced samples, but showed accumulation with the imposition of submergence stress. In case of one dimensional gel electrophoresis of soluble protein from shoot tissue of Swarna (Sub 1) in 24-96 h stress, there was alteration of five different proteins with 290, 250, 138, 121 and 101 kDa molecular weights where as in case of Ratna, there was alteration of three different proteins with 215, 138 and 121 kDa molecular weights. These proteins showed a near-regular pattern of accumulation with the continuation of submergence stress conditions.

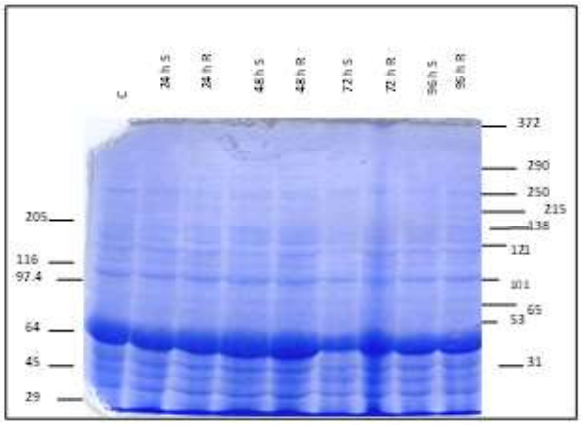


Fig 1 : Shoot protein changes in Swarna (Sub 1) rice seedlings in response to 24 h, 48 h, 72 h and 96 h of submergence stress followed by 24 h, 48 h and 72 h of recovery.

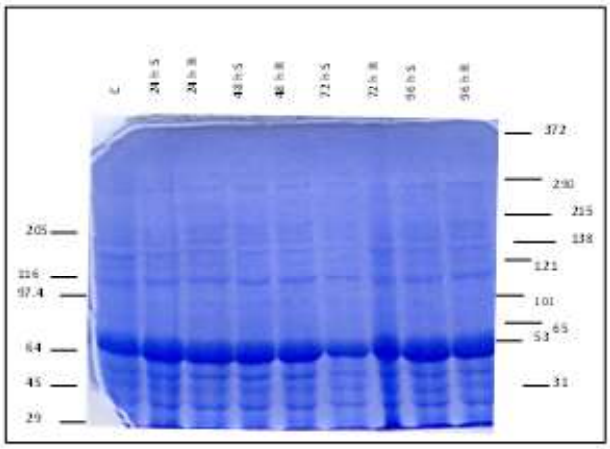


Fig 2 : Shoot protein changes in Ratna rice seedlings in response to 24 h, 48 h, 72 h and 96 h of drought stress followed by 24 h, 48 h and 72 h of recovery.

Moons *et al* (1998) noted that complete submergence of rice seedlings for 60 h increases the accumulation of a 97 kDa protein in roots. **Sachs *et al.* (1980)** analyzed proteins induced in response to anaerobic stress in maize primary roots and reported that a small number of "anaerobic polypeptides (ANPs)" accounts for more than 70% of total protein synthesis after 5 h of anaerobic stress. **Dubey *et al*., 2003** constructed proteome maps of flood-sensitive IR 54 and flood tolerant FR 13A rice types corresponding to several time points during their response to O_2 deprivation stress and identified several up- and down-regulated proteins by 1D and 2D protein gel electrophoresis followed by silver staining.

The understanding of flooding tolerance response in plants is far from complete. The protein alterations can be looked at with a viewpoint that both the induced and suppressed proteins may be important in cellular adaptation to submergence stress. The induced proteins may be the result of increased transcription, while those repressed may be due to decreased transcription. It is important to continue making attempts to unveil the mechanism of flooding tolerance response through analysis of genes/ proteins involved in this response through the power of functional genomics and proteomics-based methods. The identities of the altered stress proteins can possibly be worked out by obtaining information on the amino acid sequences of the short peptides of these proteins. The future attempts towards improving capacity of flooding tolerance through transgenic tools can possibly be further addressed if increased numbers of genes that are involved in flooding tolerance are pyramided. A lot of work needs to be carried out at the physiological, biochemical as well as molecular level in order to find the answers to the question posed by the phenomenon as complex as submergence stress.

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